STEP 1: Import Required Libraries

```
import warnings
warnings.filterwarnings('ignore')

import tensorflow as tf
from tensorflow.keras.applications import VGG16, ResNet50, MobileNet
from tensorflow.keras.preprocessing import image
import numpy as np
import matplotlib.pyplot as plt
```

STEP 2:Import Preprocessing Functions

```
from tensorflow.keras.applications.vgg16 import preprocess_input as vgg_preprocess
from tensorflow.keras.applications.resnet50 import preprocess_input as resnet_preprocess
from tensorflow.keras.applications.mobilenet import preprocess_input as mobilenet_preprocess
```

STEP 3: Load Pretrained Models

```
vgg16_model = VGG16(weights='imagenet')
resnet50_model = ResNet50(weights='imagenet')
mobilenet_model = MobileNet(weights='imagenet')
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/vgg16/vgg16_wei
553467096/553467096 ━━━━━━━━━━━━━━━━━━━━ 8s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50
102967424/102967424 ━━━━━━━━━━━━━━━━━━━━ 1s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet/mobil
17225924/17225924 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

STEP 4: Upload Image in Colab

```
from google.colab import files

uploaded = files.upload()
image_path = list(uploaded.keys())[0]
```

Choose files im1.jpg

im1.jpg(image/jpeg) - 8803 bytes, last modified: 24/03/2026 - 100% done

```
Saving im1.jpg to im1.jpg
```

STEP 5: Load and Preprocess Image

```
img = image.load_img(image_path, target_size=(224, 224))
img_array = image.img_to_array(img)
img_array = np.expand_dims(img_array, axis=0)
```

STEP 6: Apply Model-Specific Preprocessing

```
vgg_input = vgg_preprocess(img_array.copy())
resnet_input = resnet_preprocess(img_array.copy())
mobilenet_input = mobilenet_preprocess(img_array.copy())
```

STEP 7: Make Predictions Using Models

```
vgg_preds = vgg16_model.predict(vgg_input)
resnet_preds = resnet50_model.predict(resnet_input)
mobilenet_preds = mobilenet_model.predict(mobilenet_input)
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 894ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 2s 2s/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 506ms/step
```

Double-click (or enter) to edit

```
plt.imshow(img)
plt.axis('off')
plt.title("Input Image")
plt.show()
```

Input Image

STEP 8: Decode Predictions (Human Readable Output)

```
from tensorflow.keras.applications.imagenet_utils import decode_predictions

print("VGG16 Prediction:", decode_predictions(vgg_preds, top=1))
print("ResNet50 Prediction:", decode_predictions(resnet_preds, top=1))
print("MobileNet Prediction:", decode_predictions(mobilenet_preds, top=1))
```

```
Downloading data from https://storage.googleapis.com/download.tensorflow.org/data/imagenet_class_i
35363/35363 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
VGG16 Prediction: [[('n04285008', 'sports_car', np.float32(0.36326075))]]
ResNet50 Prediction: [[('n04037443', 'racer', np.float32(0.44077563))]]
MobileNet Prediction: [[('n04285008', 'sports_car', np.float32(0.9910557))]]
```

Start coding or generate with AI.